

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical
Practical – IV

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4. Business Intelligence Tools for Data Analysis:

a. Utilize a business intelligence tool (e.g., Tableau, Power BI) to extract insights from a dataset and create interactive visualizations for effective data analysis.

Dataset: netflix.csv

Aim

To perform data visualization and analysis on the Netflix dataset using Tableau by creating different charts, applying filters, developing an interactive dashboard, and identifying important insights.

A. Connect and Understand the Data

1. Connecting the Dataset

The netflix.csv dataset was connected to Tableau using the **Text File** connection option. After loading the dataset, all available fields and records were displayed in Tableau.

Output:

The screenshot shows the Tableau interface with the 'netflix_titles.csv' dataset connected. The 'Connections' pane on the left shows 'netflix_titles' selected. The 'Files' pane shows 'house_price_re...on_dataset.csv' and 'netflix_titles.csv'. The 'Fields' pane lists fields: Show ID, Type, Title, Director, and Cast. The main view shows a table of data with columns: Show ID, Type, Title, Director, Cast, and Country.

2. Understanding the Columns and Data Types

The dataset contains information about Netflix titles such as Show ID, Type, Title, Director, Cast, Country, Date Added, Release Year, Rating, Duration, Genre, and Description.

The fields were examined to identify text, numeric, and date data types.

- **Text fields:** Show ID, Type, Title, Director, Cast, Country, Rating, Duration, Genres, and Description.
- **Numeric field:** Release Year.
- **Date field:** Date Added.

Output:

netflix_titles.csv		14 fields 8807 rows		100
Name netflix_titles.csv				
Description				
No description available.				
Fields				
Type	Field Name	Physical Table	Remote Field N...	
Abc	Show Id	netflix_titles.csv	show_id	
Abc	Type	netflix_titles.csv	type	
Abc	Title	netflix_titles.csv	title	
Abc	Director	netflix_titles.csv	director	
Abc	Cast	netflix_titles.csv	cast	
	Country	netflix_titles.csv	country	
	date_added	netflix_titles.csv	date_added	
	Release Year	netflix_titles.csv	release_year	
Abc	Rating	netflix_titles.csv	rating	
Abc	Duration	netflix_titles.csv	duration	
Abc	Listed In	netflix_titles.csv	listed_in	
Abc	Description	netflix_titles.csv	description	
-#	Movie Durati	Calculation	Calculation_063...	
.ili.	Movie Durati	Bin	Movie Duration (...)	

3. Finding Total Records

The total number of records was calculated by counting the show_id field.

Output:

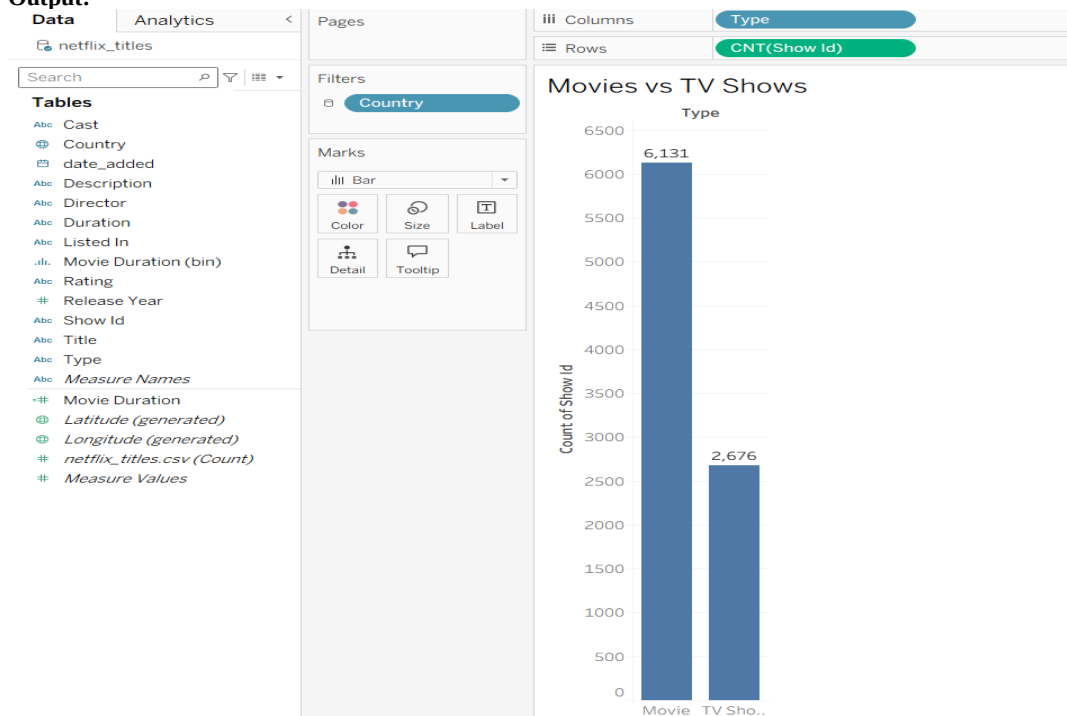
netflix_titles.csv	14 fields 8807 rows
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Observation: The dataset contains 7,787 total Netflix titles.

4. Finding the Number of Movies and TV Shows

The Type field was used with the count of show_id to determine the total number of Movies and TV Shows available in the dataset.

Output:



Observation:

Movies: 5,377

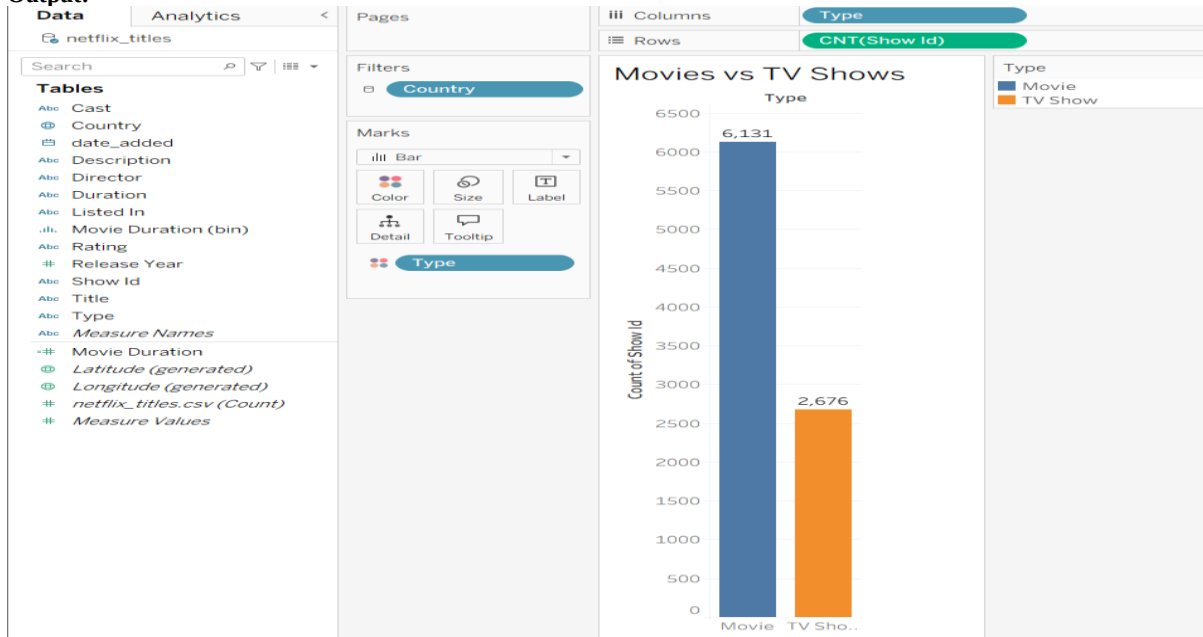
TV Shows: 2,410

B. Create Different Visualizations:

1. Bar Chart – Movies vs TV Shows

A bar chart was created using **Type** and **Count of Show ID** to compare the number of Movies and TV Shows.

Output:

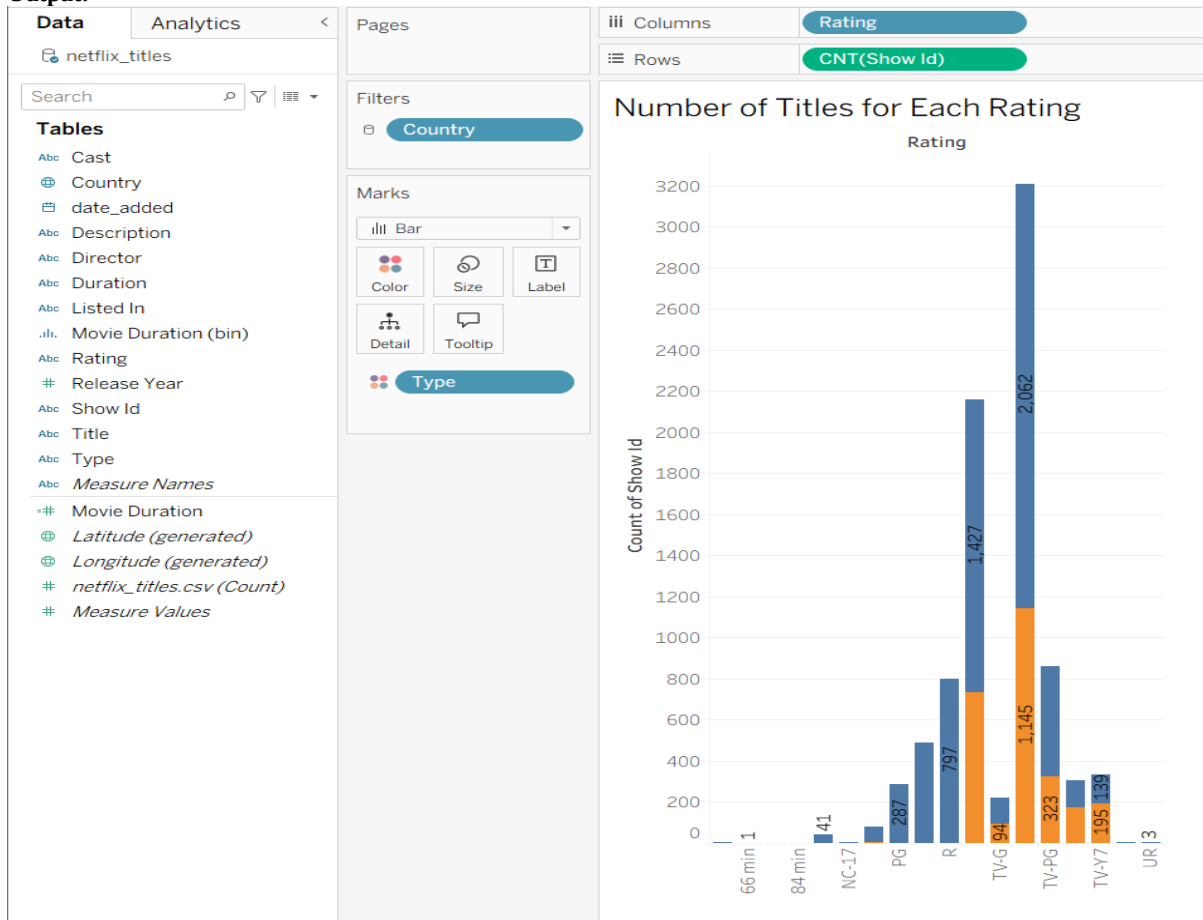


Observation: Movies have a higher number of titles than TV Shows.

2. Bar Chart – Titles by Rating

A bar chart was created using **Rating** and **Count of Show ID** to display the number of Netflix titles for each rating category.

Output:

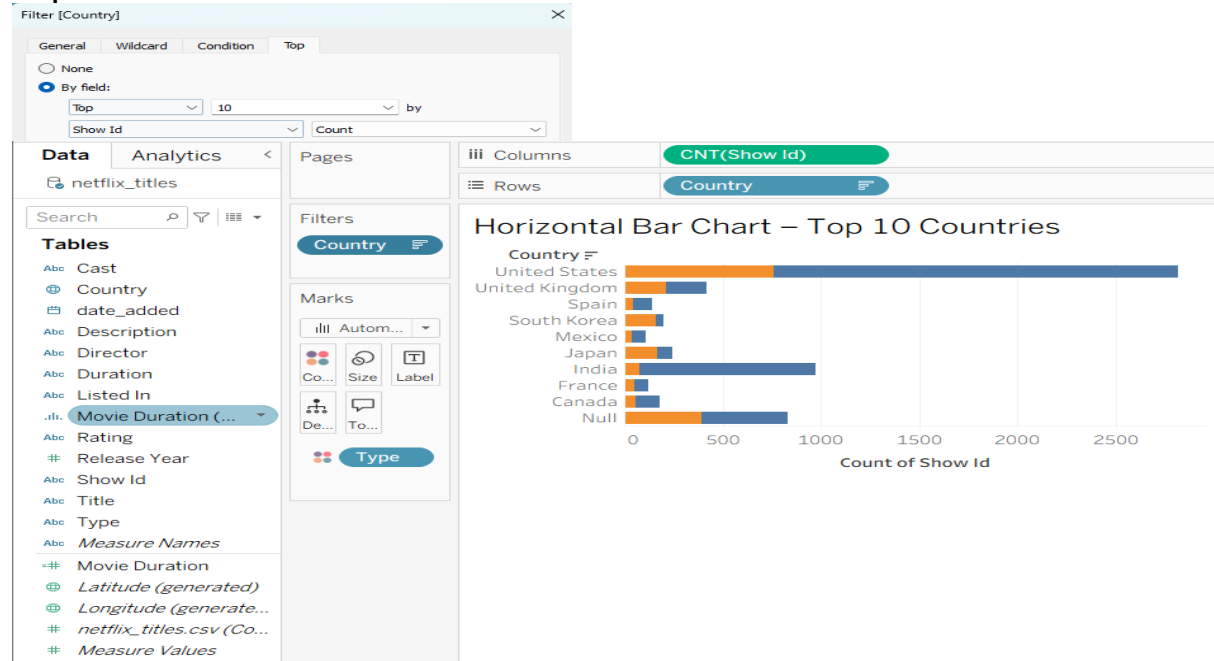


Observation: TV-MA has the highest number of titles with 2,863 titles.

3. Horizontal Bar Chart – Top 10 Countries

A horizontal bar chart was created to display the top 10 countries with the highest number of Netflix titles. The data was sorted in descending order and the Top 10 filter was applied.

Output:

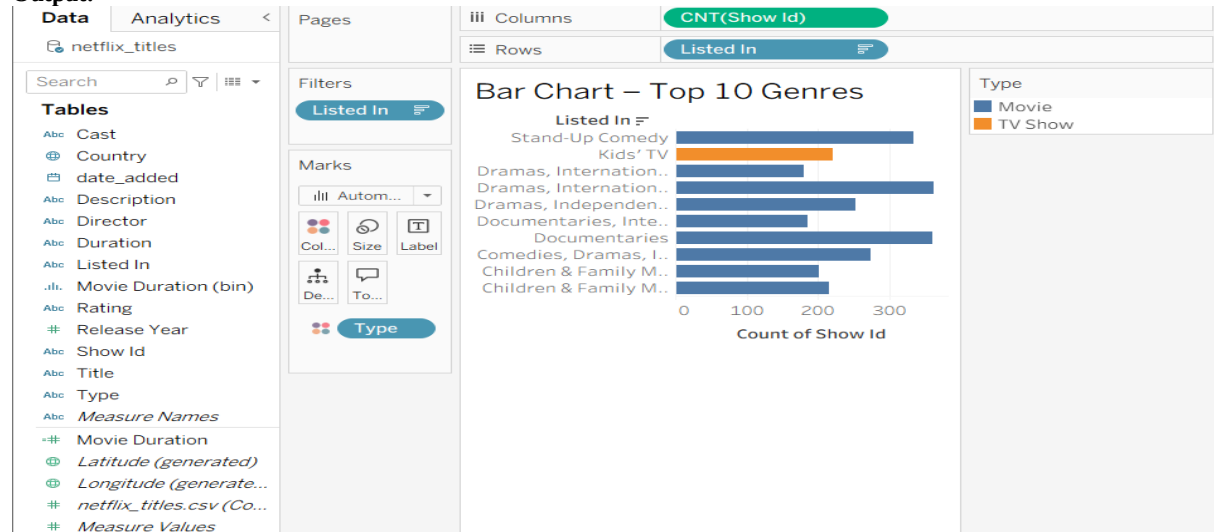


Observation: United States has the highest number of titles among the country values in the dataset, with **2,555 titles**.

4. Bar Chart – Top 10 Genres

A bar chart was created using the **Genres** field and **Count of Show ID**. The chart was sorted in descending order, and the Top 10 filter was applied.

Output:

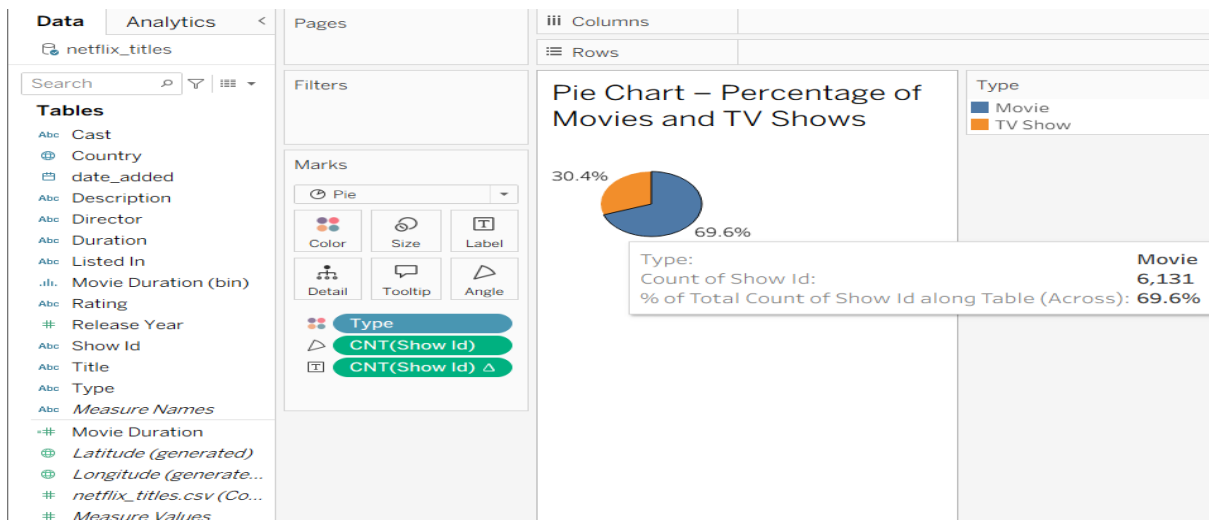


Observation: Documentaries is the highest exact genre value in the dataset, with **334 titles**.

5. Pie Chart – Movies and TV Shows Percentage

A pie chart was created using **Type** as Color and **Count of Show ID** as Angle. Percentage of total was displayed using labels.

Output:

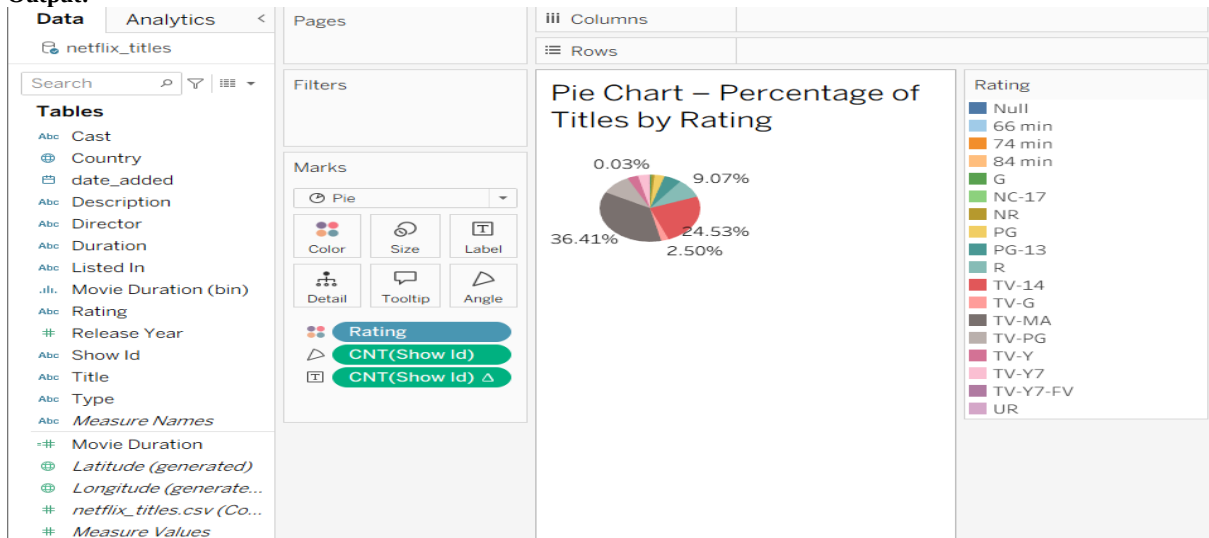


Observation: Movies represent a larger percentage of Netflix content than TV Shows.

6. Pie Chart – Rating Percentage

A pie chart was created to show the percentage distribution of Netflix titles across different rating categories.

Output:

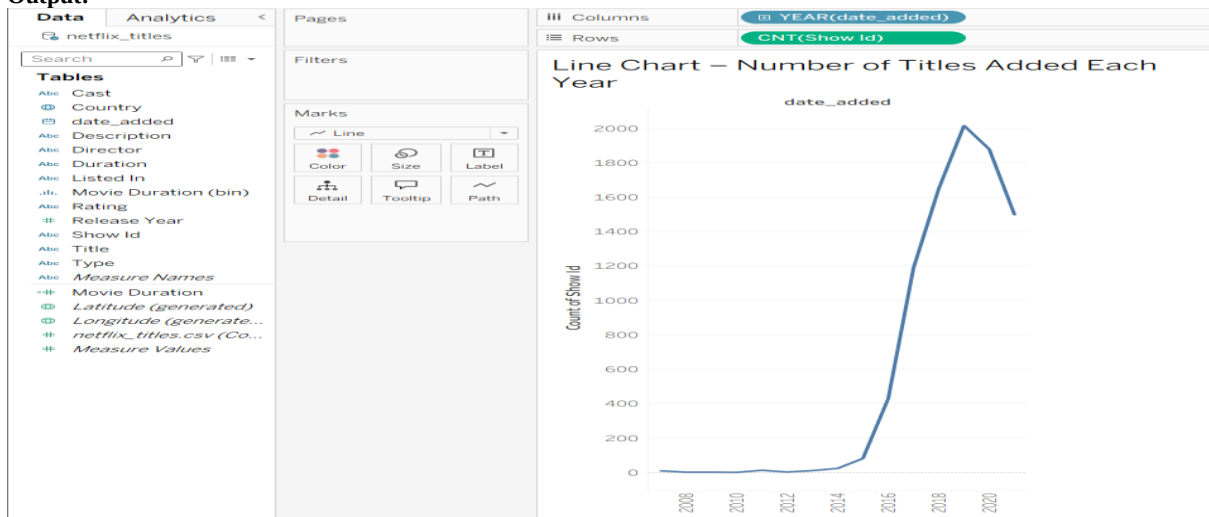


Observation: TV-MA represents the largest share among the rating categories.

7. Line Chart – Titles Added Each Year

A line chart was created using YEAR(Date Added) and Count of Show ID to analyze the number of Netflix titles added each year.

Output:

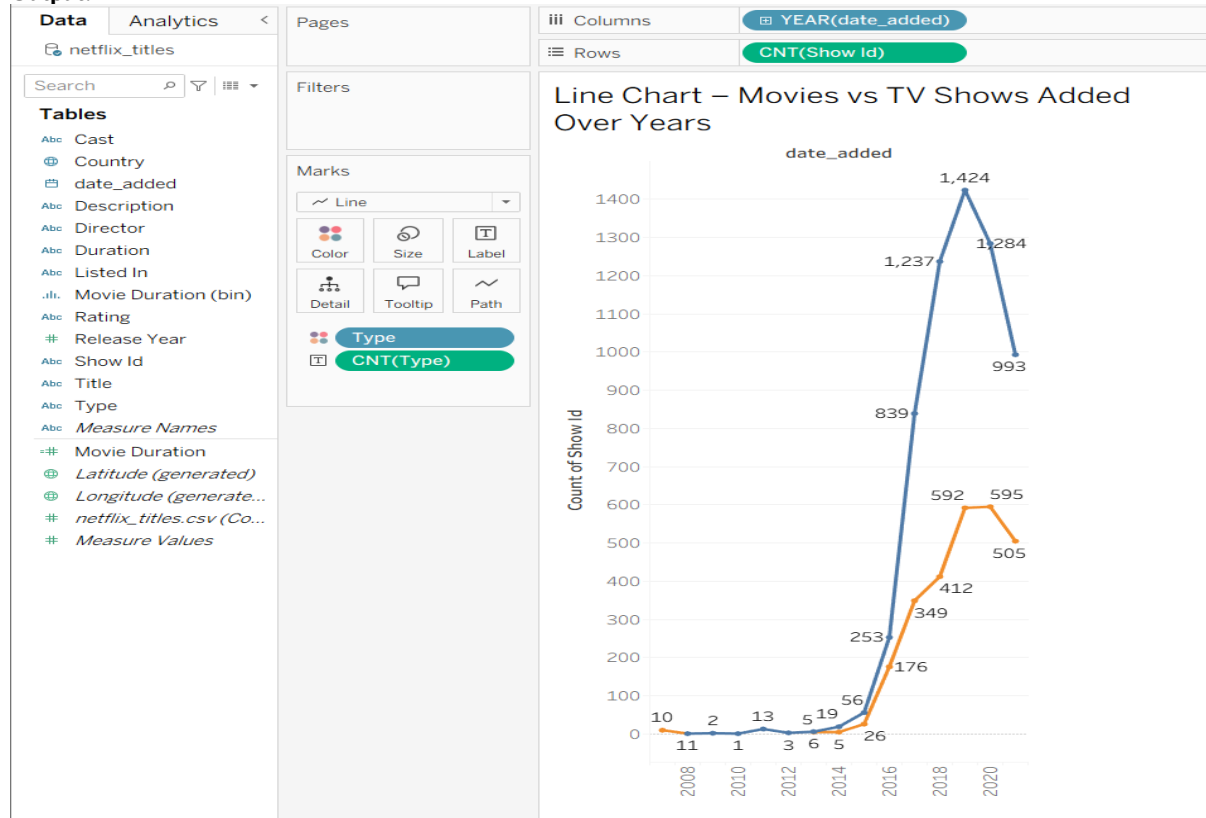


Observation: The highest number of titles was added in 2019, with 2,136 titles.

8. Line Chart – Movies and TV Shows Added Over the Years

A line chart was created using **YEAR(Date Added)**, **Count of Show ID**, and **Type** as Color to compare Movies and TV Shows added over different years.

Output:

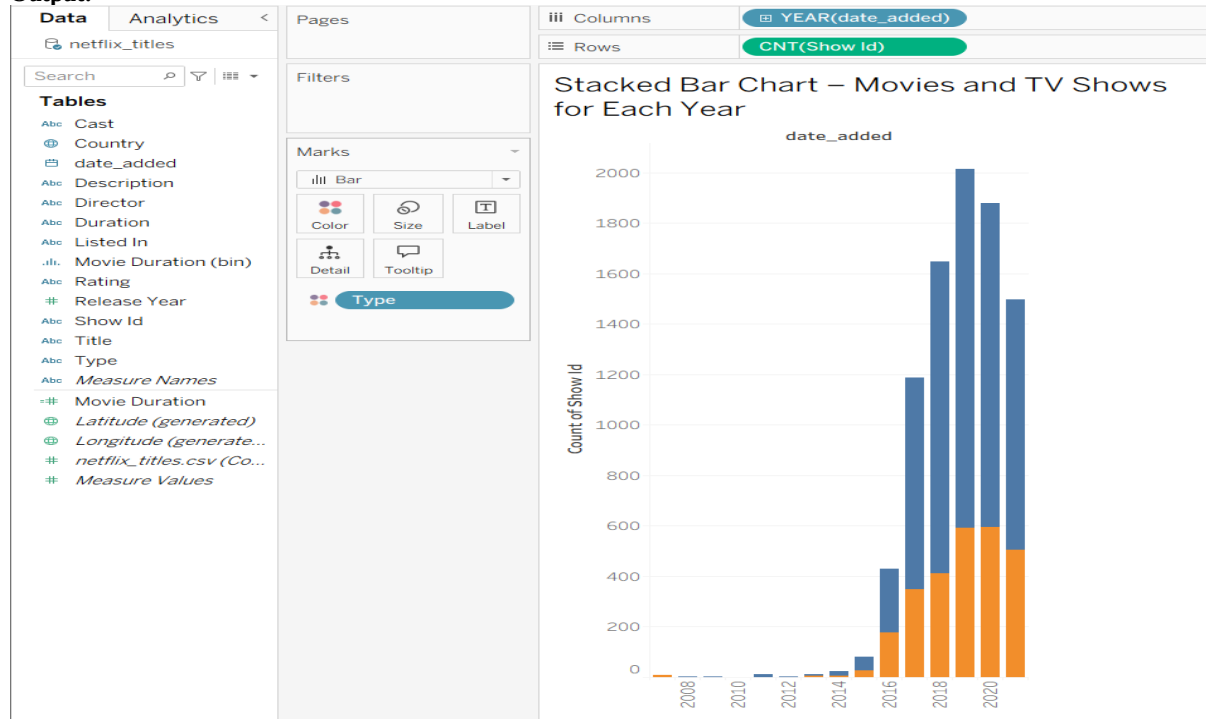


Observation: The line chart shows that Movies were added in greater numbers than TV Shows during the major years represented in the dataset.

9. Stacked Bar Chart – Movies and TV Shows by Year

A stacked bar chart was created using **YEAR(Date Added)** and **Count of Show ID**, with **Type** used as Color.

Output:

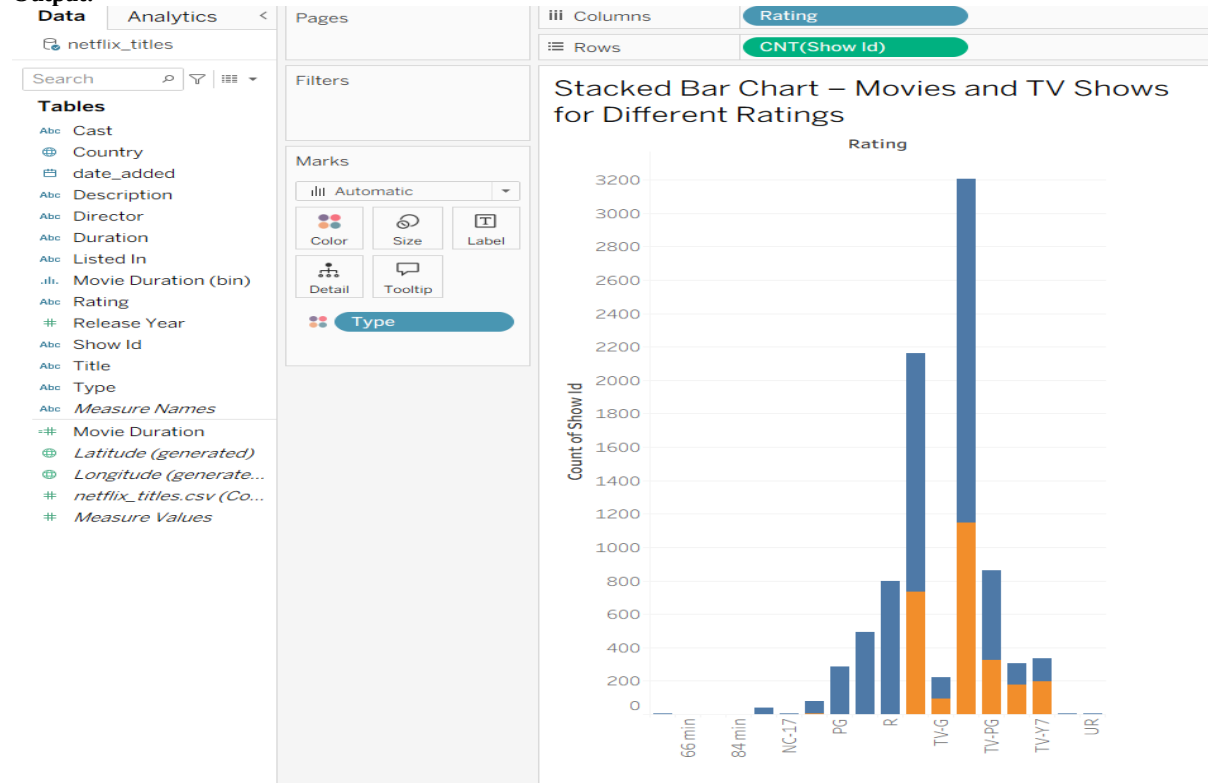


Observation: Movies form the larger portion of the stacked bars for most years.

10. Stacked Bar Chart – Movies and TV Shows by Rating

A stacked bar chart was created using **Rating** and **Count of Show ID**, with **Type** used as Color.

Output:

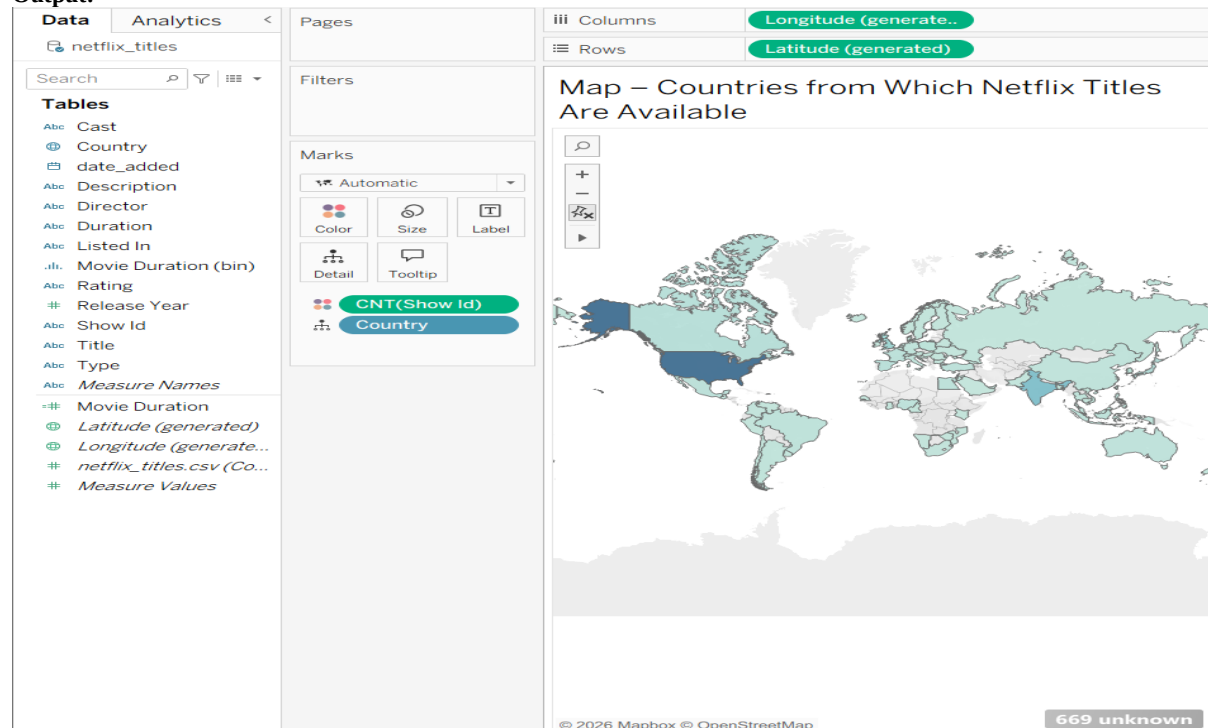


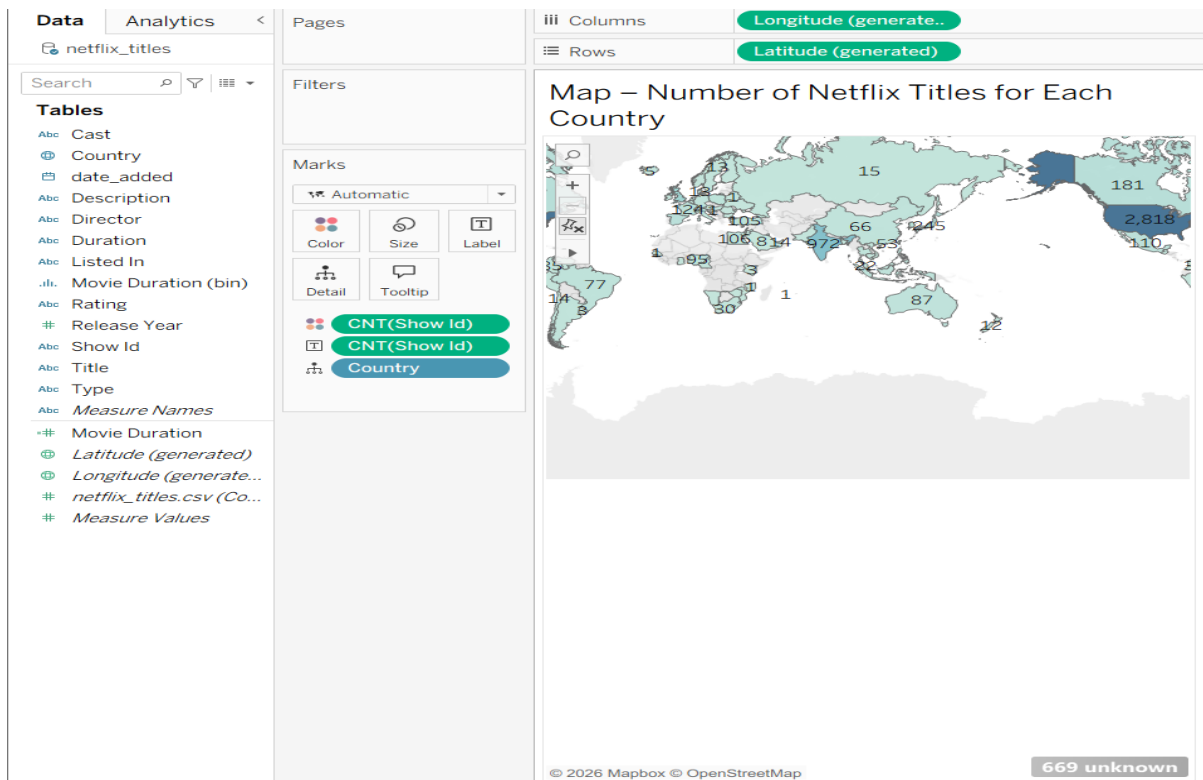
Observation: The chart shows the distribution of Movies and TV Shows across different rating categories.

11. Map – Netflix Titles by Country

A geographical map was created using the **Country** field. **Count of Show ID** was added to display the number of Netflix titles associated with each country.

Output:



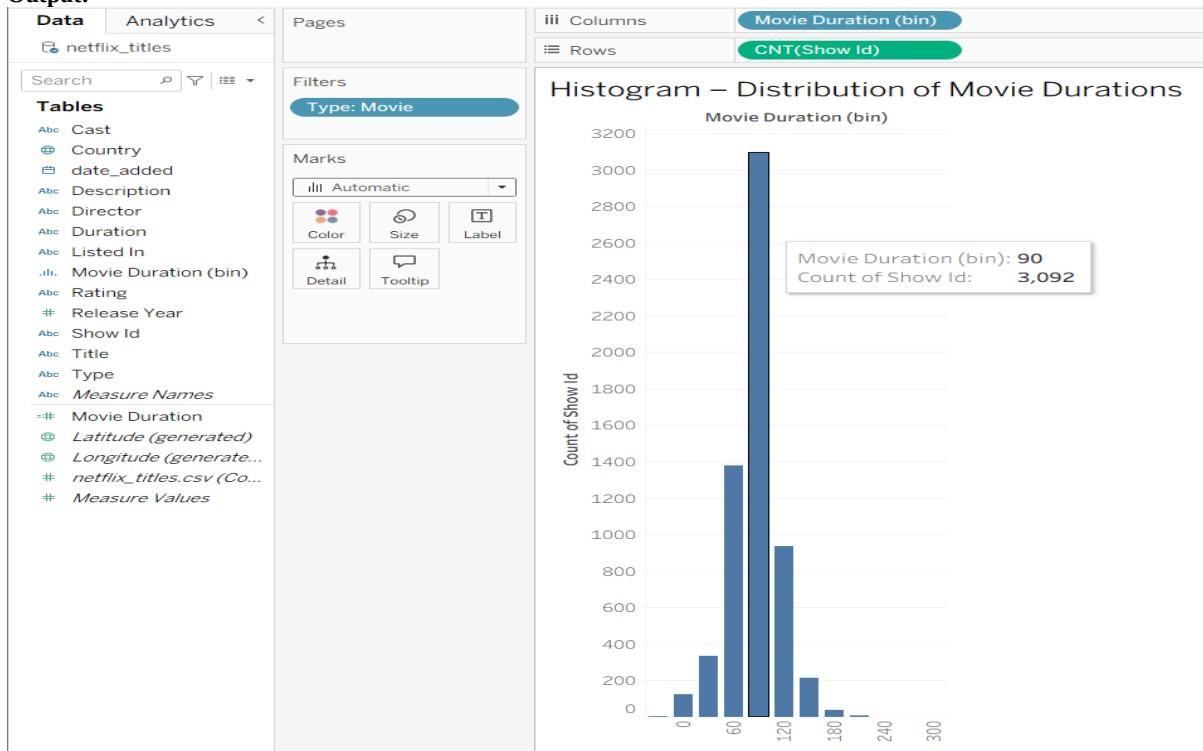


Observation: The map shows the geographical distribution of Netflix titles across different countries.

12. Histogram – Distribution of Movie Duration

The data was filtered to include only **Movies**. The numeric **Duration** field was used to create bins, and a histogram was generated to analyze the distribution of movie durations.

Output:

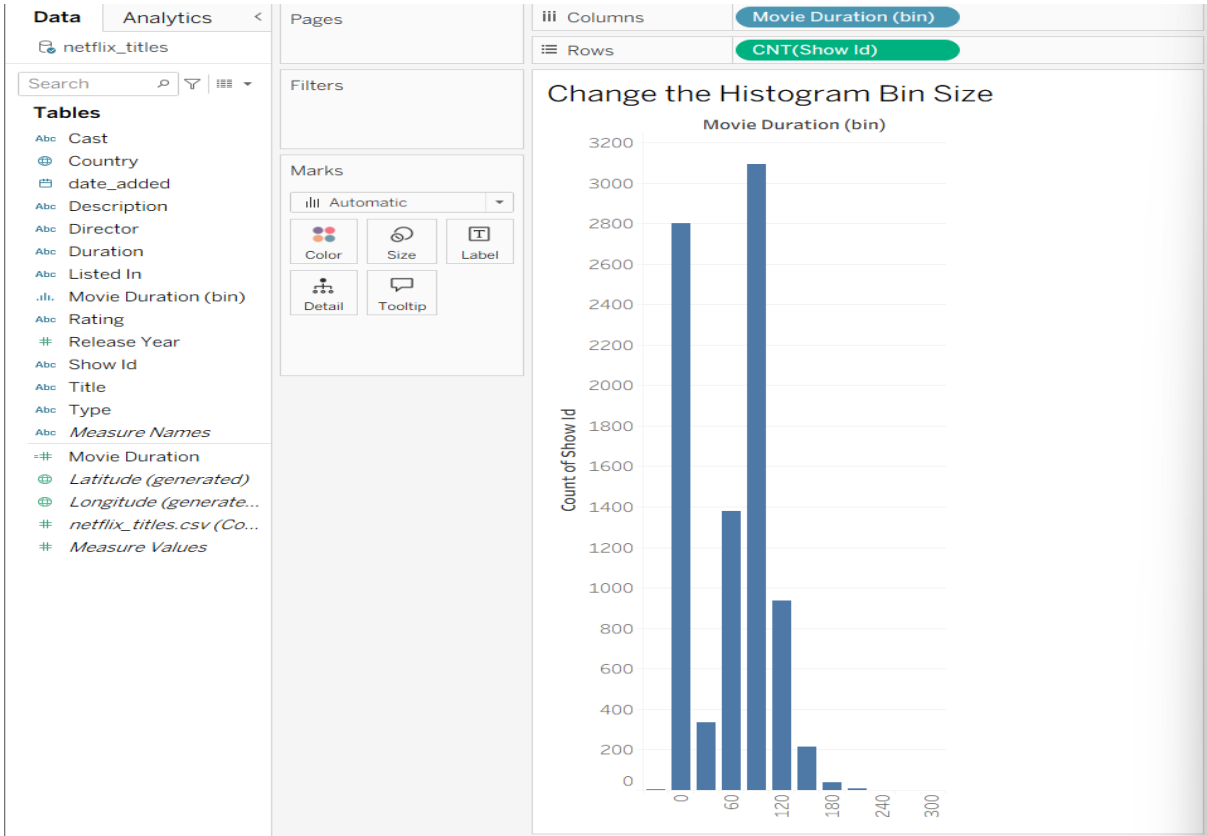


Observation: Most Movies are concentrated around the 90–120 minute duration range when using 30-minute bins.

13. Changing the Histogram Bin Size

The bin size of the Movie Duration histogram was changed to observe how different interval sizes affect the visualization.

Output:



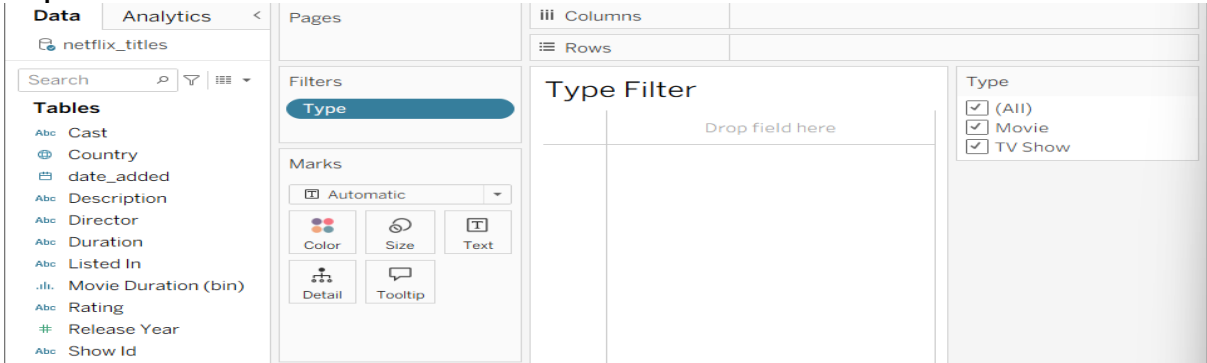
Observation: A smaller bin size provides more detailed information about movie durations, while a larger bin size provides a simpler overall distribution.

C. Apply Filters

1. Type Filter

A **Type** filter was added to allow the visualization to display only Movies or only TV Shows.

Output:

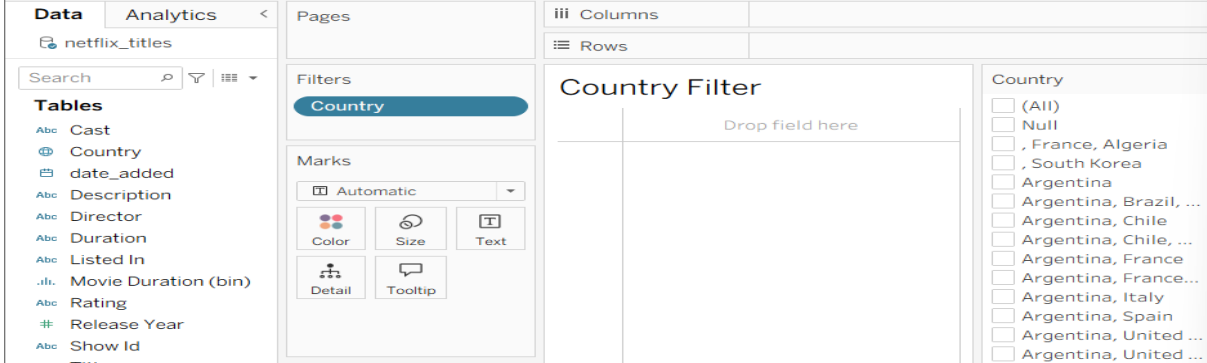


Observation: Selecting Movie or TV Show changes the visualization according to the selected content type.

2. Country Filter

A **Country** filter was added to display Netflix titles from a selected country.

Output:

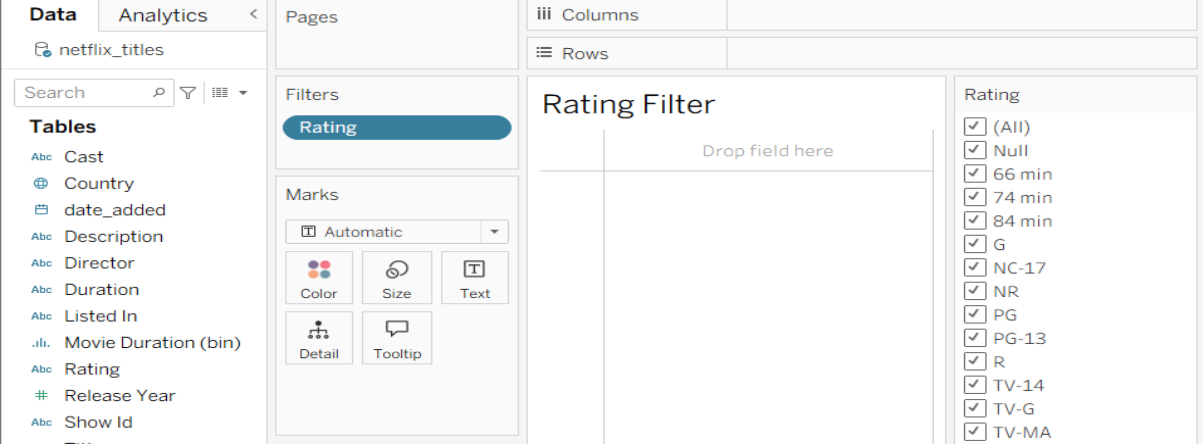


Observation: The chart updates according to the selected country.

3. Rating Filter

A **Rating** filter was added to display Netflix titles according to selected rating categories.

Output:

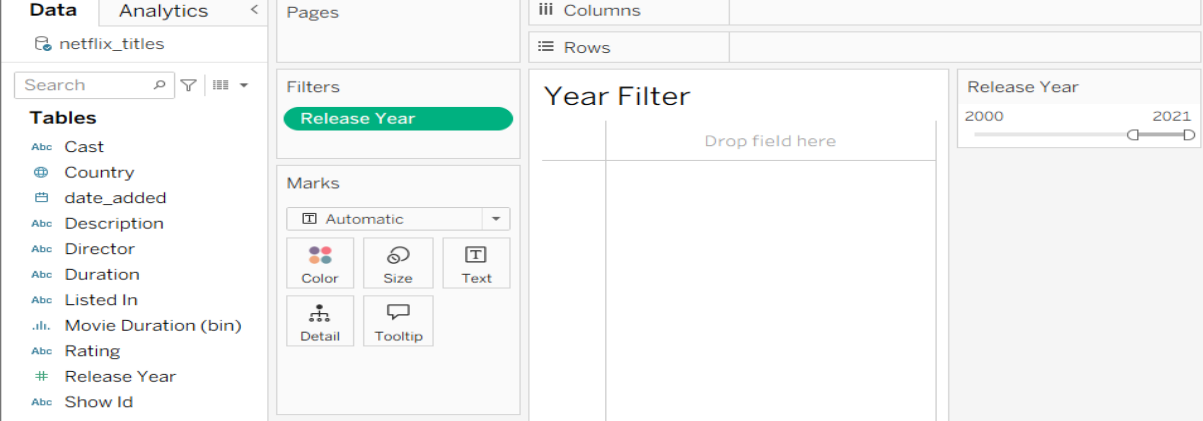


Observation: The visualization changes according to the selected rating.

4. Year Filter

A **Year** filter was added using the appropriate year field to display titles from a selected year.

Output:

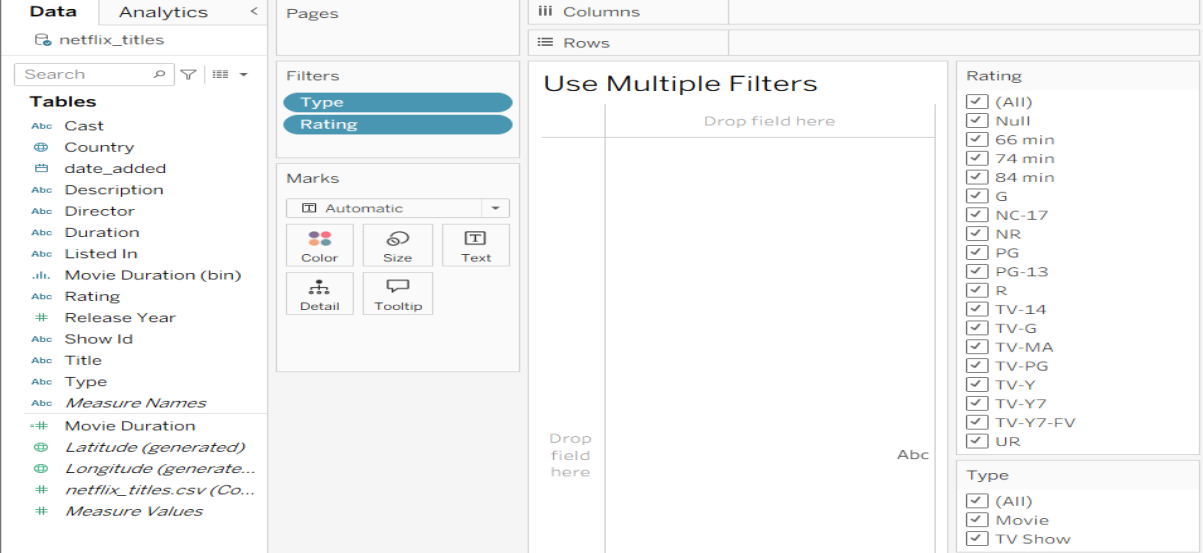


Observation: The visualization displays data only for the selected year.

5. Applying Multiple Filters

Two or more filters such as **Type, Country, Rating, and Year** were applied together to perform more detailed analysis of Netflix content.

Output:



Observation: Multiple filters allow users to analyze specific groups of Netflix titles more effectively.

D. Create a Dashboard

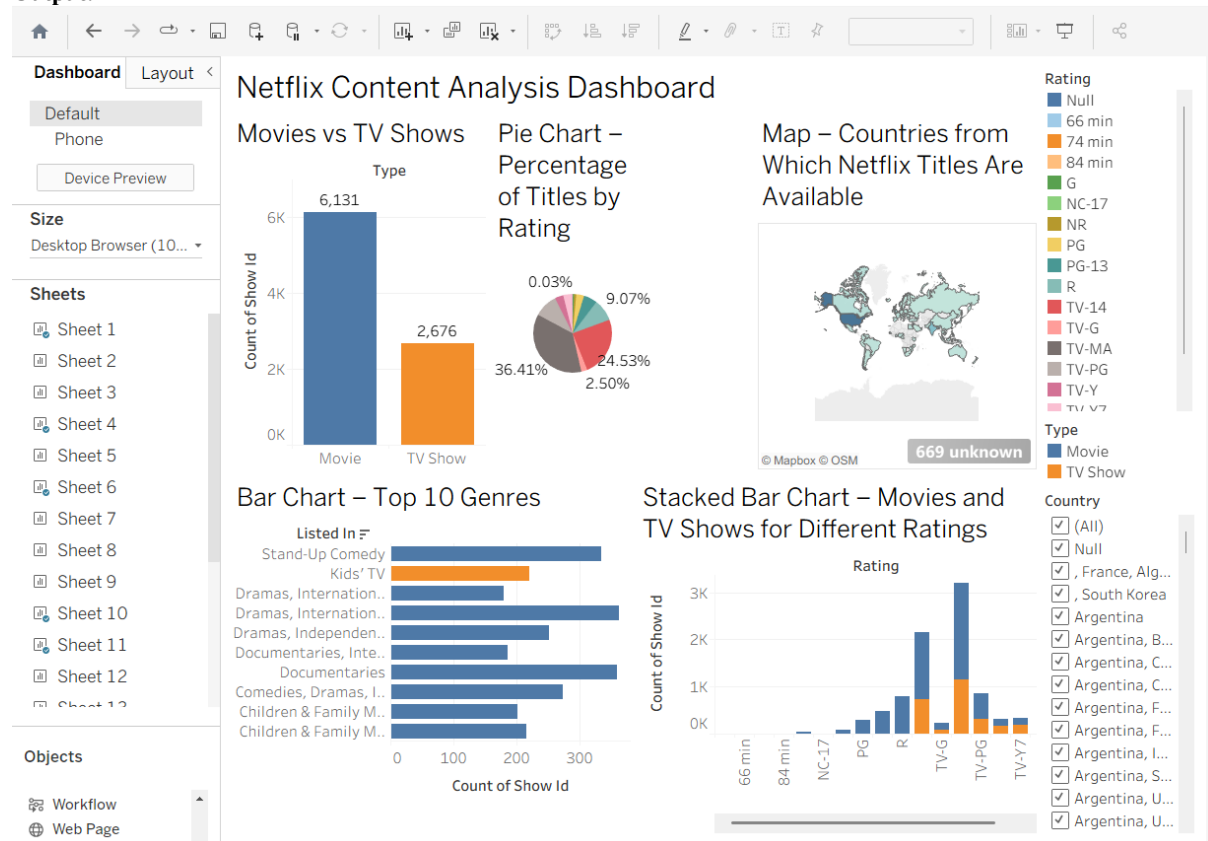
A new Tableau Dashboard was created by combining the important visualizations developed in the previous steps. The charts were arranged properly to make the dashboard clear and easy to understand.

Interactive filters such as **Type, Country, Rating, and Year** were added to allow users to explore the data dynamically. The dashboard was given the title:

Netflix Content Analysis Dashboard

Suitable chart titles, labels, and filters were added to improve readability and usability.

Output:



Dashboard Description

The dashboard combines different visualizations to provide an overall analysis of Netflix content. It allows users to compare Movies and TV Shows, analyze ratings and genres, identify countries with more titles, study yearly trends, view geographical distribution, and examine Movie duration patterns. Interactive filters make the dashboard dynamic and user-friendly.

E. Insights from the Data

1. Which type of content is available more on Netflix?

Answer: Movies are available more on Netflix, with 5,377 titles compared with 2,410 TV Shows.

2. Which country has the highest number of Netflix titles?

Answer: United States has the highest number of titles among the country values in the dataset, with 2,555 titles.

3. Which Rating has the highest number of titles?

Answer: TV-MA has the highest number of Netflix titles, with 2,863 titles.

4. Which year had the highest number of titles added to Netflix?

Answer: The highest number of titles was added in 2019, with 2,136 titles.

5. Which Genre has the highest number of titles?

Answer: Documentaries is the highest exact genre value in the dataset, with 334 titles.

6. Which Movie duration range has the highest number of Movies?

Answer: The 90–120 minutes range has the highest concentration of Movies when using 30-minute duration bins.

Five Important Observations

1. **Movies** have a higher number of titles than TV Shows in the Netflix dataset.
2. **United States** has the highest number of titles among the country values in the dataset.
3. **TV-MA** is the rating category with the highest number of titles.

4. **2019** had the highest number of titles added to Netflix, with **2,136 titles**.
 5. The Movie Duration histogram shows that many Movies are concentrated around the **90–120 minute** range.
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Conclusions

Conclusion 1

Netflix contains a large collection of Movies and TV Shows, with Movies representing the larger portion of the dataset.

Conclusion 2

The analysis shows that content varies across countries, ratings, genres, and years, with some categories having significantly more titles than others.

Conclusion 3

Tableau visualizations such as bar charts, pie charts, line charts, stacked bar charts, maps, and histograms make it easier to understand Netflix content patterns and trends.

Result

The netflix.csv dataset was successfully analyzed using Tableau. Different visualizations including **bar charts, pie charts, line charts, stacked bar charts, geographical maps, and histograms** were created. Filters were applied, and an interactive **Netflix Content Analysis Dashboard** was developed successfully. Important insights regarding content type, countries, ratings, genres, yearly trends, and movie durations were identified.